FUZZY LOGIC CONTROL

Dr. D. CHAKRADHAR
Associate Professor

Dept. of Mechanical Engineering

IIT PALAKKAD

Nurturing minds for a better world

Fuzzy logic controllers are used in a wide range of applications, including:

Automatic washing machines: The controller adjusts the wash cycle based on the amount of dirt and stains on the clothes.

Heating, Ventilating, Air Conditioning (HVAC) systems: The controller adjusts the temperature and airflow based on the room's temperature and the desired temperature set by the user.

Automatic cruise control for vehicles: The controller maintains a safe distance from the vehicle in front and adjusts the speed accordingly.

Industrial process control: The controller adjusts various process variables such as temperature, pressure, and flow rate to maintain optimal conditions for the production process.

Robotics: Fuzzy logic controllers are used in robots to control their movements and actions based on sensor inputs.

Power systems: Fuzzy logic controllers are used in power systems to control the generation and distribution of electric power in an efficient and reliable way.

- What is Fuzzy Logic?
- Boolean Logic Vs Fuzzy Logic
- Basic concepts in Fuzzy Systems.
- Fuzzy controller block diagram.
- Fuzzy Logic System
- Membership Functions

What is Fuzzy ?

Fuzzy – “not clear, distinct, or precise; blurred”

A form of knowledge representation suitable for notions that cannot be defined precisely but depend upon their contexts.

Boolean Logic

Boolean Logic is restricted to **“0”** and **“1”**

If the colour is red then “1”

If the colour is not red then “0”

Fuzzy Logic

Fuzzy Logic values varies between “0” and “1”

Slowest
Speed = 0.0 - 0.25

Slow
Speed = 0.25 – 0.50

Fast
Speed = 0.50 – 0.75

Fastest
Speed = 0.75 – 1.00

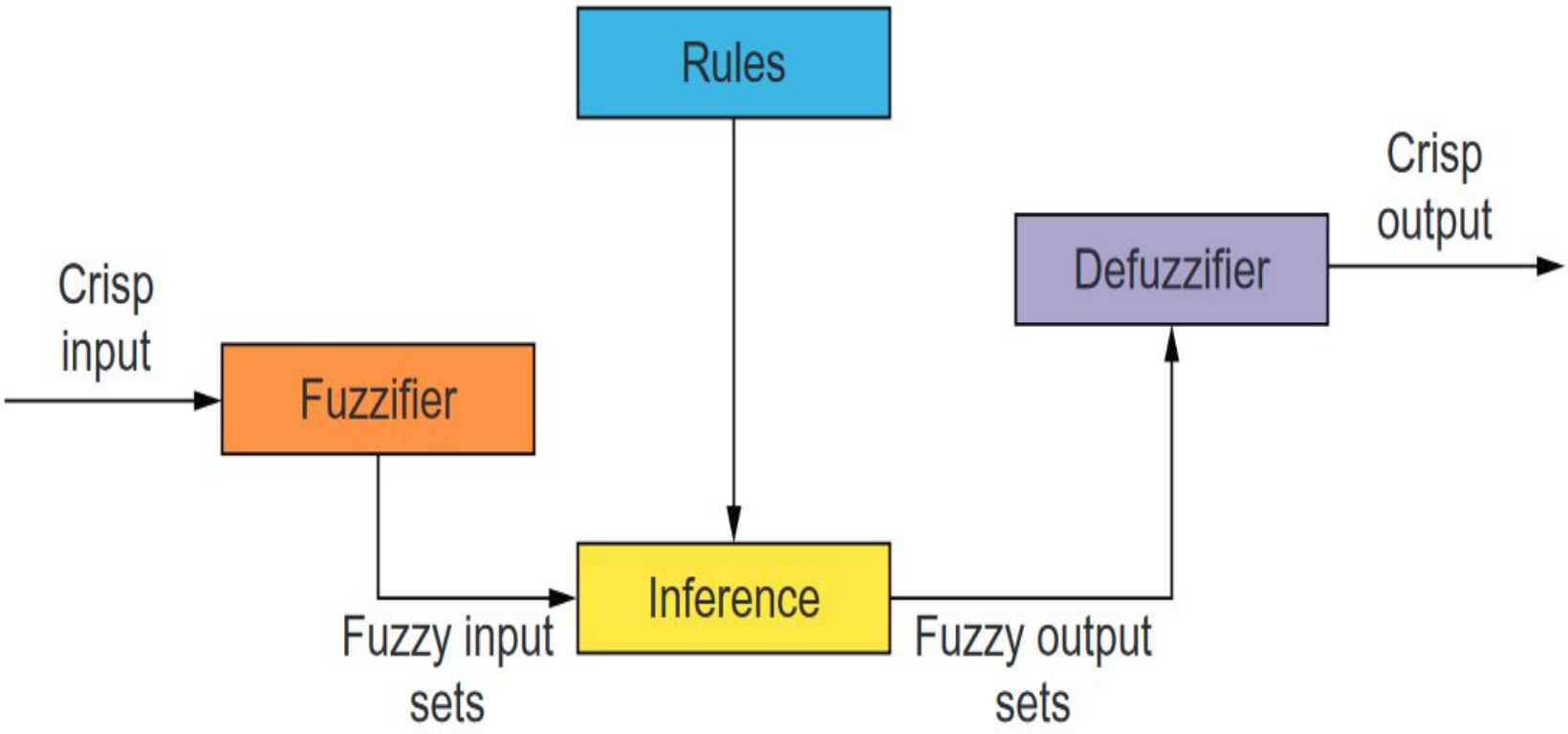
Basic concepts in Fuzzy systems

- **Membership function (MF)** - A function that specifies the degree to which a given input belongs to a set.

- **Degree of membership-** The output of a membership function, this value is always limited to between 0 and 1. Also known as a membership value or membership grade.

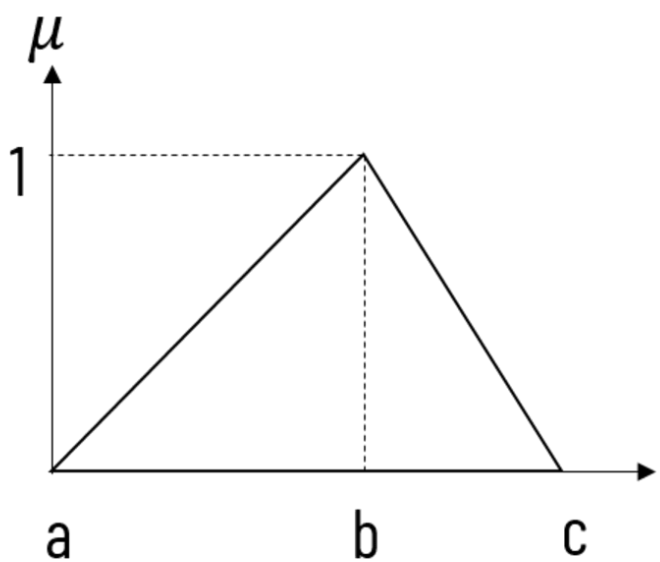
- **Membership functions** are used in the fuzzification and defuzzification steps of an FLS (fuzzy logic system), to map the non-fuzzy input values to fuzzy linguistic terms and vice versa.

Fuzzy controller block Diagram



Membership Functions

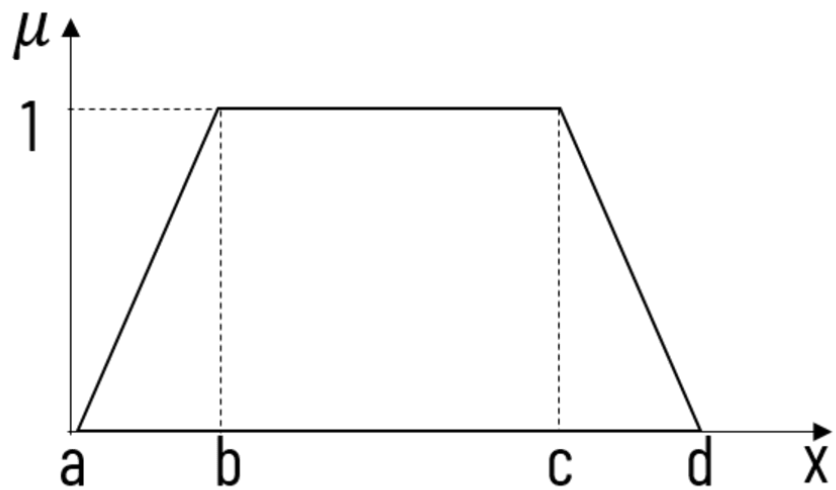
Triangular membership function:



$$\mu_{triangle}(x;a,b,c) = \begin{cases} 0, & x \le a \\ \dfrac{x-a}{b-a}, & a \le x \le b \\ \dfrac{c-x}{c-b}, & b \le x \le c \\ 0, & c \le x \end{cases}$$

$$= \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$

Trapezoidal membership function:

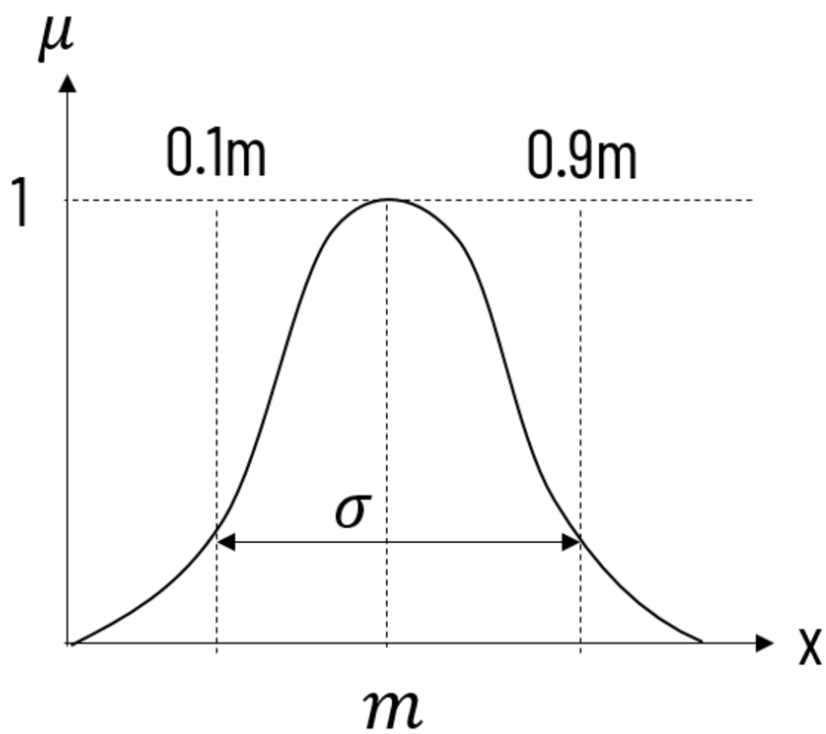


$$\mu_{trapezoidal}(x; a, b, c, d) = \begin{cases} 0, & x \leq a \\ \dfrac{x-a}{b-a}, & a \leq x \leq b \\ 1, & b \leq x \leq c \\ \dfrac{d-x}{d-c}, & c \leq x \leq d \\ 0, & d \leq x \end{cases}$$

$$= \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right)$$

Gaussian membership function:

A Gaussian MF is specified by two parameters {m, σ}. In this function, m represents the mean / center of the gaussian curve and σ represents the spread of the curve.



$$\mu_{gaussian}(x; m, \sigma) = e^{-\frac{1}{2}\left(\frac{x-m}{\sigma}\right)^2}$$

Restaurant example for fuzzy logic

Create a fuzzy control system which models how to tip at a restaurant (from 0 to 25 %) based on the service (rated in points from 0 to 10) and food quality (0 to 10).

Inputs

1. Service

Linguistic variables : Poor, average, good

Range : 0 to 10

2. Food quality

Linguistic variables : Poor, average, good

Range : 0 to 10

Output

1. Tip %

Linguistic variables : Low, medium, high

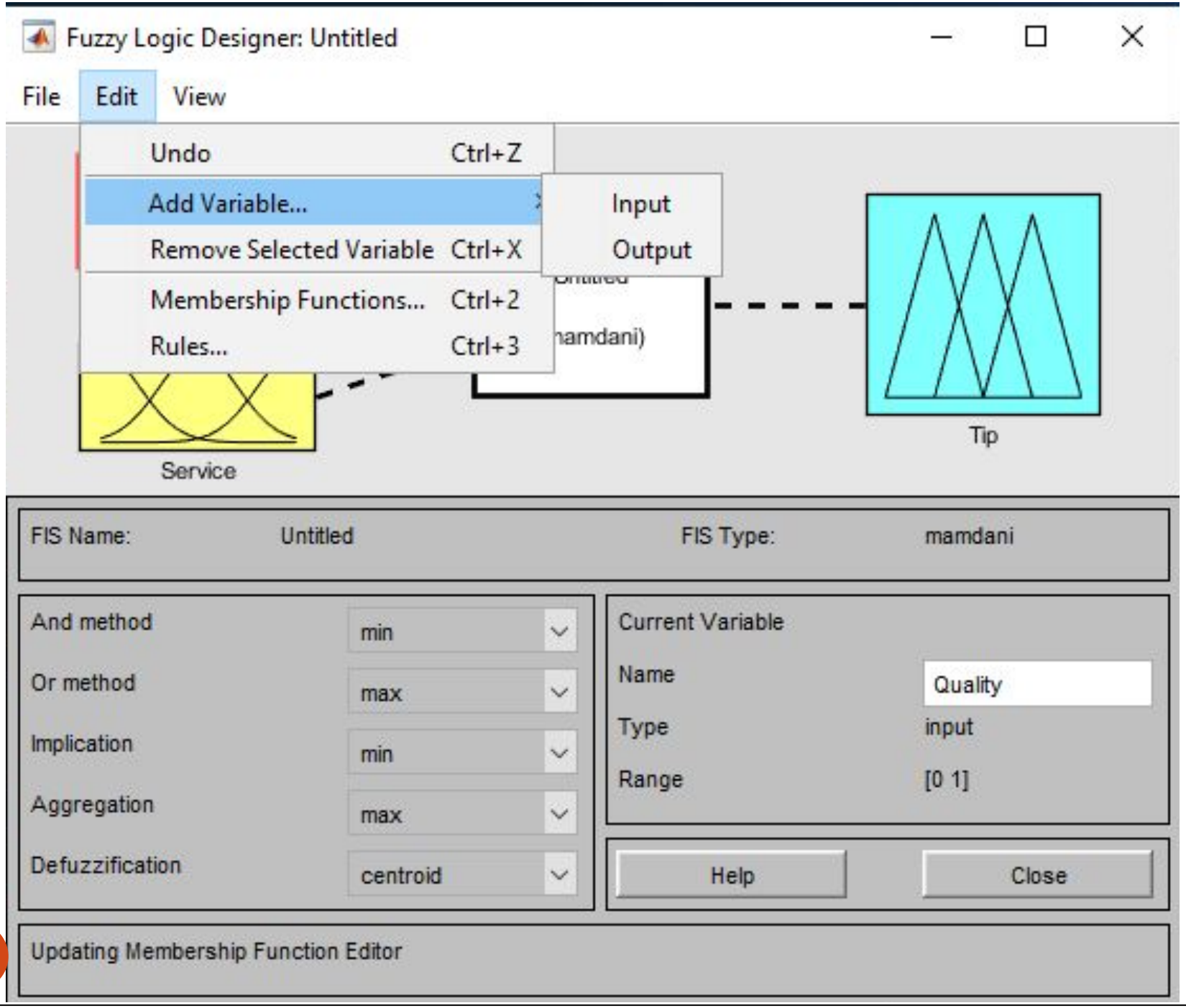
Range : 0 to 25

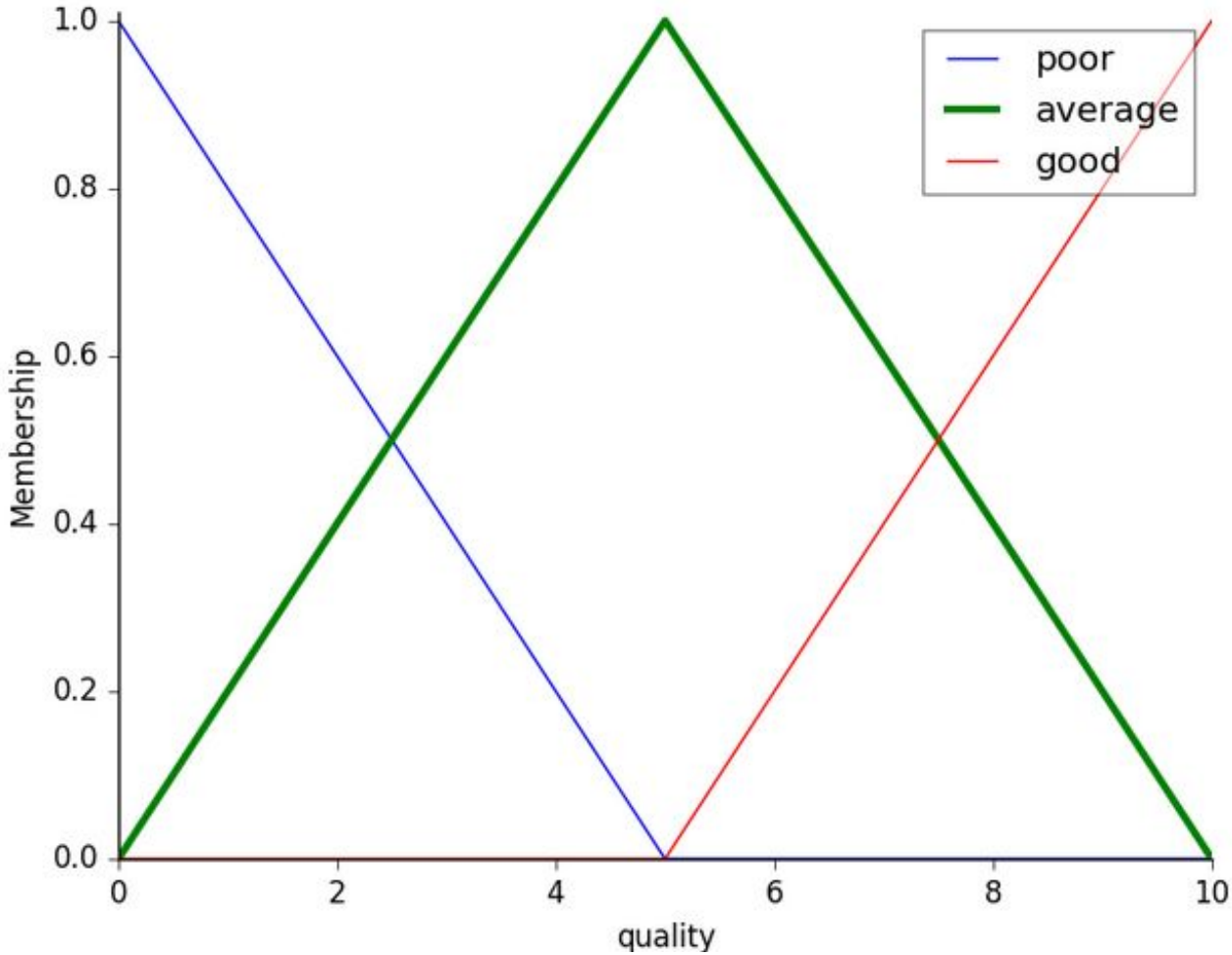
Rules

1. **If** the food quality is **poor** **OR** the service is **poor**, then the tip will be **low**

2. If the service is **average**, then the tip will be **medium**

3. If the food quality is **good** **OR** the service is **good,** then the tip will be **high**.


Fuzzy Logic Designer: Untitled
File
Edit
View
Undo Ctrl+Z
Add Variable...
Remove Selected Variable Ctrl+X
Membership Functions... Ctrl+2
Rules... Ctrl+3
Input
Output
Service
Tip
FIS Name: Untitled
FIS Type: mamdani
And method min
Or method max
Implication min
Aggregation max
Defuzzification centroid
Current Variable
Name Quality
Type input
Range [0 1]
Help
Close
Updating Membership Function Editor



Triangular membership function for food quality

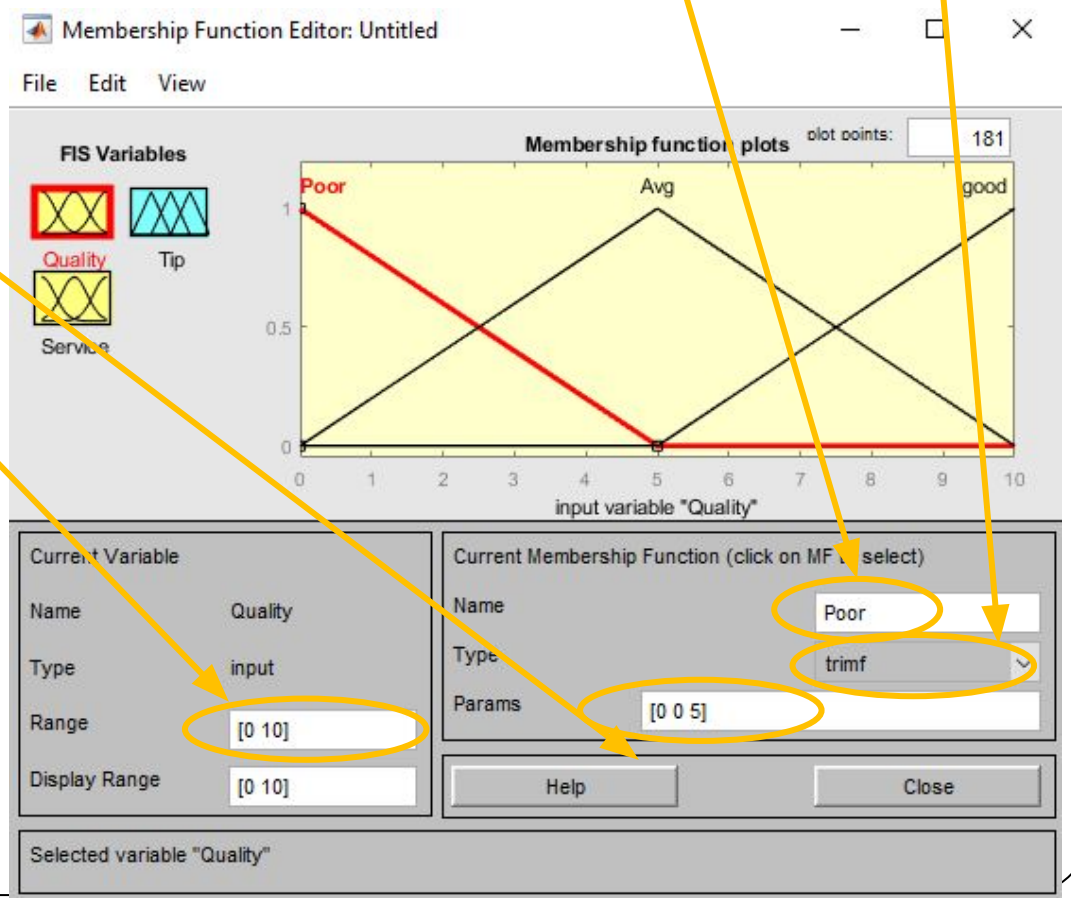
Defining membership functions for each variables

Select membership function shape

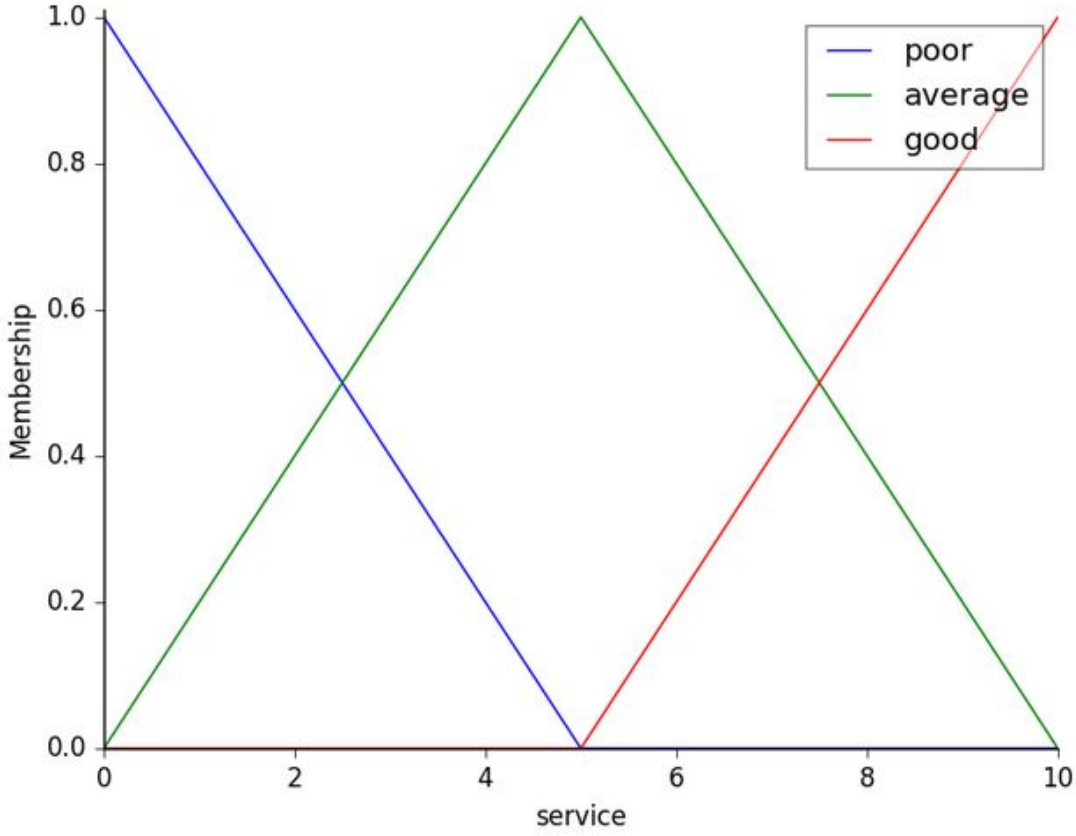
Rename the linguistic variables

Adjust the shape using parameters

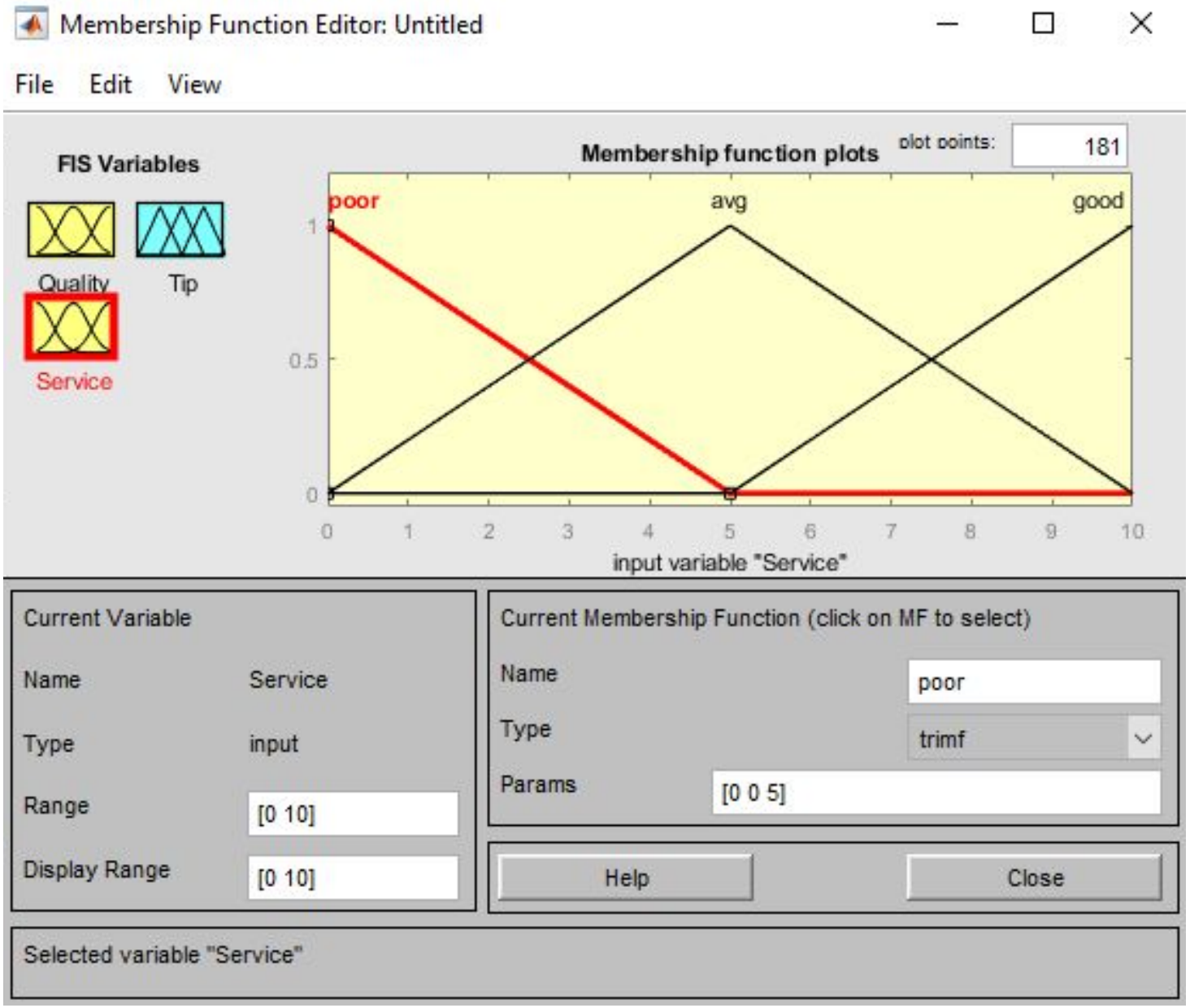
- **Select variable range**



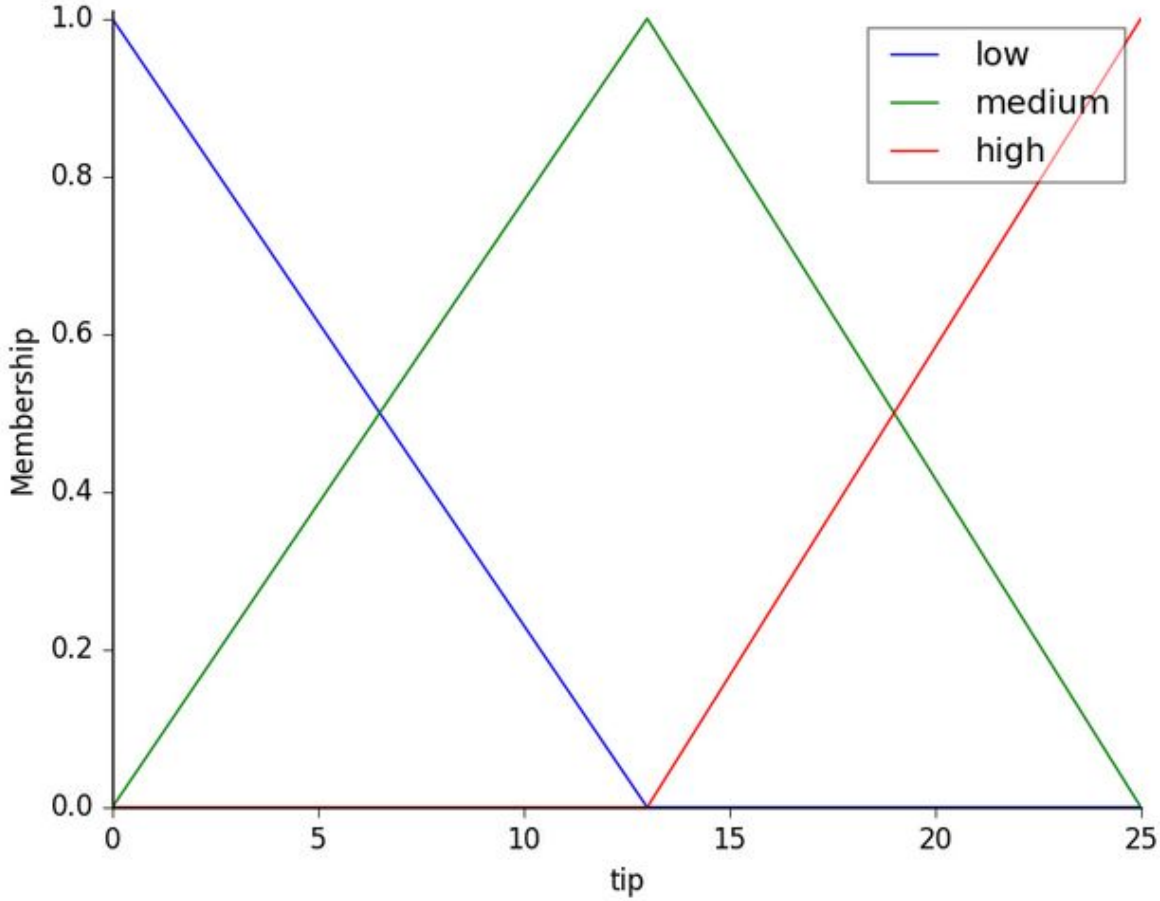
Triangular membership function for food quality



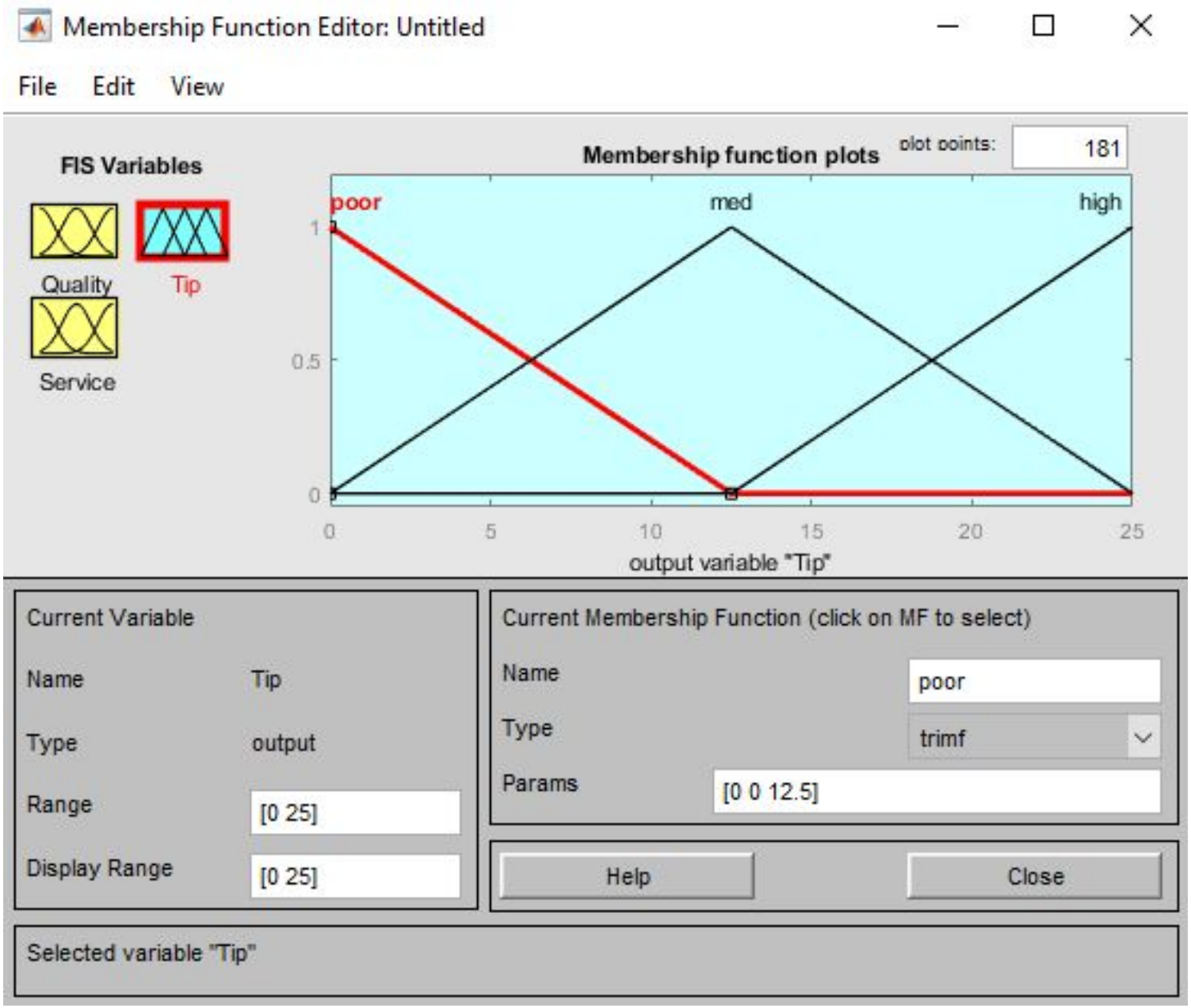
Triangular membership function for service



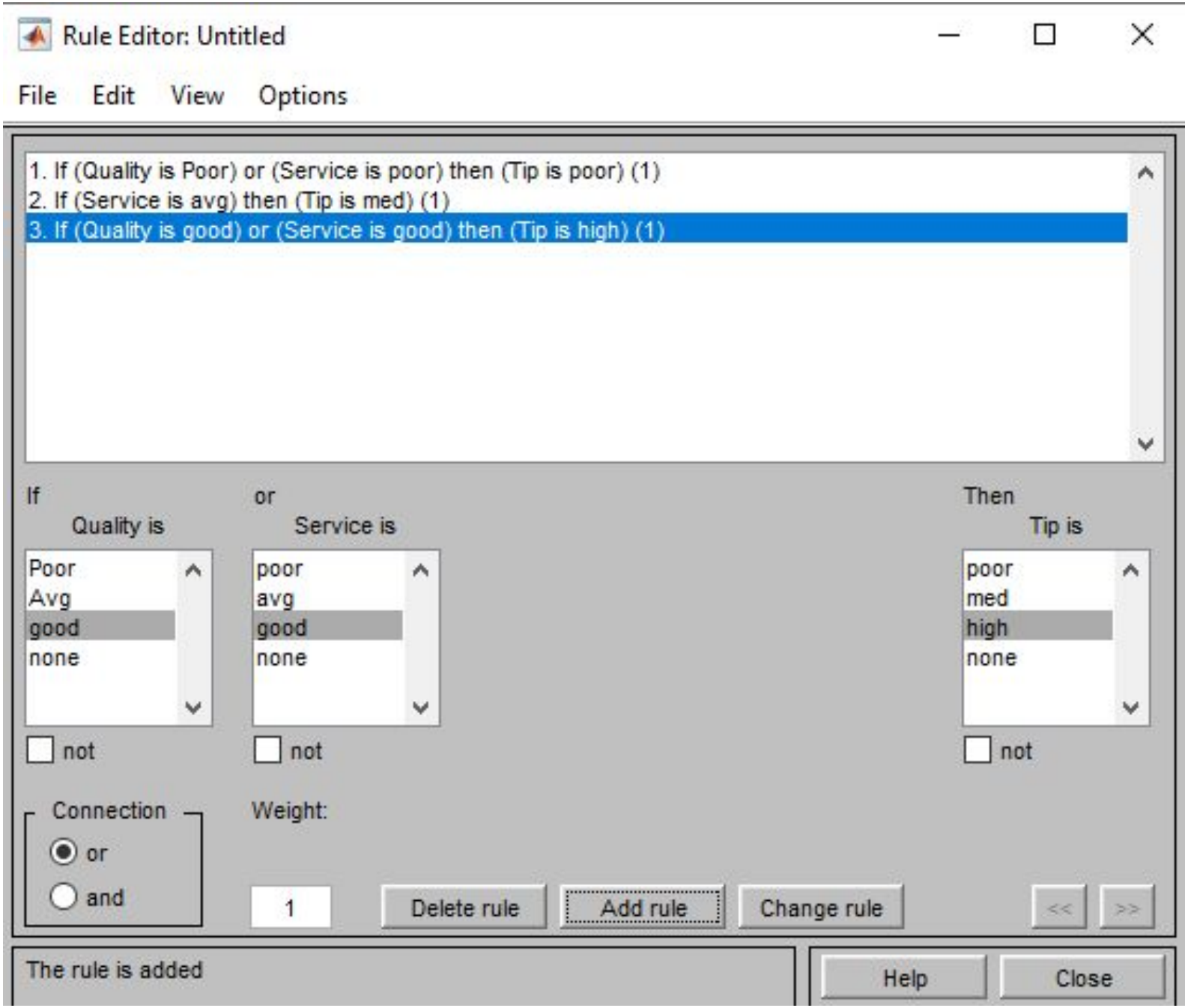
Defining the membership function variables for service



Triangular membership function for tip



Defining the membership function variables for tip


Rule Editor: Untitled
File
Edit
View
Options
1. If (Quality is Poor) or (Service is poor) then (Tip is poor) (1)
2. If (Service is avg) then (Tip is med) (1)
3. If (Quality is good) or (Service is good) then (Tip is high) (1)
If
Quality is
Poor
Avg
good
none
not
or
Service is
poor
avg
good
none
not
Then
Tip is
poor
med
high
none
not
Connection
or
and
Weight:
1
Delete rule
Add rule
Change rule
<<
>>
The rule is added
Help
Close

Rule Viewer: Untitled
File
Edit
View
Options
Quality = 6.5
Service = 9.8
Tip = 19.9
1
2
3
0
10
0
10
0
25
Input:
[6.5 9.8]
Plot points:
101
Move:
left
right
down
up
Opened system Untitled, 3 rules
Help
Close

THANK YOU